

Uncertainty and Sentiment in ARK Company Filings

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github.com/victor4628/FRE7871-NLP

1. What I did

I study 1,536 filings from 91 companies in the ARK holdings universe, submitted during 2021-2025. I count negative and uncertainty words, compare proportions with TF-IDF, test language trends, and relate uncertainty to subsequent stock volatility and negative language to four-day excess returns. The strongest result is rising negative language within annual reports; incremental volatility prediction and pooled return prediction are not established.

2. Data construction

Security / company screen	Removed	Remain
Unique cleaned securities in frozen snapshot	0	124
SEC ticker-to-CIK mapping found	7	117
At least one 2021-2025 10-K/10-Q	24	93
Unique companies (GOOG and GOOGL count as one)	1	92

Table 1. Sequential filing screens	Removed	Remain
Downloaded security-filing records	0	1,702
Duplicate Alphabet accessions (retain GOOGL)	20	1,682
Parsed non-amended 10-K/10-Q	0	1682
At least 2,000 words (10-K) or 1,000 words (10-Q)	0	1682
Earliest filing per company-calendar quarter	14	1668
Usable day 0 and day -1 price at least \$3	108	1560
At least 60 return days before and after day 0	23	1537
Missing filing share count needed for size control	1	1536

The last row is an additional regression-data requirement: one JOBY filing has no usable share count, so size cannot be calculated. No further filing is lost to other controls or outcomes. All exhibits use this common sample. The earliest-filing rule uses submission quarter, across form types; it removes 14 distinct 10-Qs, not duplicate text.

Day 0 follows the assignment: convert acceptance time to Eastern; shift its date forward one day at or after 16:00; take the later of that date and the filing date, then the next available trading day. This moves 860 retained filings. Share counts are read from the same filing cover or a dated same-accession fact, never a later report. Prices and shares are aligned for splits.

Parsing keeps visible inline-XBRL narrative, removes hidden resources, and strips tables with more than 15% digits among non-space characters. The uppercase tokenizer retains words of at least two characters; there is no stemming or stopword removal. No filing failed parsing.

3. Word lists

The Loughran-McDonald dictionary supplies 2,355 Negative words and 297 Uncertainty words; 40 occur in both lists. Negative words describe adverse conditions; uncertainty words qualify what may happen or what is known. Shared words and shared disclosure topics mean these are conceptually different but empirically related measures.

4. Method

For each filing, the proportional measure is list-word occurrences divided by all tokens. TF-IDF downweights repetition and words that occur in many documents. For term i in document j :

$$w_{ij} = [(1 + \ln tf_{ij}) / (1 + \ln a_j)] \ln(N / df_j)$$

The weight is zero if tf is zero. Here tf is the term count, a is total tokens divided by distinct tokens, N is the number of documents and df is the number containing the term. A category score sums these weights. Natural logs reproduce the instructor checks: 0.8480, 0.2885, 0.5026. IDF is recomputed within each pooled or form-specific regression corpus.

For trends, each company receives equal weight within form and filing quarter. Separate 10-K and 10-Q series remove the seasonal form-mix problem. Aggregate regressions use time and quarter-of-year effects; Newey-West standard errors with four lags allow serial correlation. The within-company model adds company effects, comparing changes inside the same company. Company-clustered errors allow its repeated filings to be related.

For market outcomes, each language measure enters as a continuous regressor. Reported proportional-score coefficients are effects per 1 percentage-point increase in words (for example, 1.8% to 2.8%); reported TF-IDF coefficients are effects per 1 TF-IDF unit. The estimation code standardizes scores internally, then rescales the coefficients and standard errors to these original units. This does not change fitted values, t-statistics or p-values.

Both models include log size (day -1 price times filing shares), log mean daily dollar volume and stock-minus-SPY buy-and-hold return over [-60,-6], report type, company effects and calendar-quarter effects. Table 5 adds log prior volatility only in its second specification; Table 6 includes it. The code uses a 10-Q dummy with 10-K as reference, algebraically equivalent to a 10-K dummy with the intercept changed.

Realized volatility is daily-return sample SD multiplied by $\sqrt{252}$: 55 returns on [-60,-6] before and 60 on [+4,+63] after. The four-day return compounds days [0,+3] from the day -1 close, then subtracts the corresponding SPY return. No missing returns are filled. Outcome errors cluster by company.

5. What the measures are made of

Table 2. Filing-level summary statistics, by report type

Form / measure	N	Mean	SD	P25	Median	P75
10-K / Negative %	381	2.318	0.413	2.045	2.344	2.582
10-K / Negative TF-IDF	381	163.515	56.188	119.102	149.910	199.653
10-K / Uncertainty %	381	1.935	0.241	1.790	1.953	2.114
10-K / Uncertainty TF-IDF	381	26.685	8.484	20.720	25.281	31.680
10-Q / Negative %	1,155	2.108	0.940	1.266	1.890	2.993
10-Q / Negative TF-IDF	1,155	79.247	65.775	24.322	55.471	127.759
10-Q / Uncertainty %	1,155	1.808	0.582	1.317	1.644	2.405
10-Q / Uncertainty TF-IDF	1,155	13.471	7.275	7.874	11.925	18.467

Proportions are percentages of all words; TF-IDF is the sum of term weights. P25 and P75 are percentiles.

Table 3. Thirty most frequent words on each list

Each share is the word's fraction of all occurrences on its own list.

Rank	Negative word	List share	Uncertainty word	List share
1	LOSS	5.16%	MAY	34.79%
2	ADVERSELY	4.08%	COULD	18.35%
3	LOSSES	3.18%	RISK	5.05%
4	CLAIMS	3.01%	RISKS	4.53%
5	ADVERSE	2.72%	BELIEVE	2.95%
6	AGAINST	2.42%	APPROXIMATELY	2.65%
7	LITIGATION	1.94%	ASSUMPTIONS	1.91%
8	UNABLE	1.92%	INTANGIBLE	1.70%
9	HARM	1.86%	POSSIBLE	1.32%
10	FAILURE	1.79%	UNCERTAINTIES	1.16%
11	FAIL	1.32%	FLUCTUATIONS	1.14%
12	IMPAIRMENT	1.24%	MIGHT	1.09%
13	NEGATIVELY	1.14%	UNCERTAIN	1.03%
14	DIFFICULT	1.13%	ANTICIPATED	0.98%
15	PENALTIES	1.13%	VOLATILITY	0.94%
16	CRITICAL	1.02%	DEPEND	0.94%
17	NEGATIVE	1.02%	PREDICT	0.94%
18	DAMAGES	0.92%	UNCERTAINTY	0.89%
19	DECLINE	0.91%	DIFFER	0.86%
20	DELAYS	0.91%	ANTICIPATE	0.80%
21	RESTATED	0.89%	EXPOSURE	0.79%
22	LIMITATIONS	0.86%	CONTINGENT	0.78%
23	DELAY	0.81%	VARIABLE	0.71%
24	VOLATILITY	0.77%	DEPENDS	0.68%
25	CLOSING	0.76%	PENDING	0.65%
26	CHALLENGES	0.76%	DEPENDENT	0.58%
27	FINES	0.71%	CONTINGENCIES	0.56%
28	HARMED	0.69%	PROBABLE	0.52%
29	DISRUPTIONS	0.69%	ASSUMED	0.52%
30	INVESTIGATIONS	0.67%	VARY	0.49%

Q1. What is each measure actually made of? The top ten words contribute 28.1% of Negative and 74.4% of Uncertainty occurrences. Uncertainty therefore measures much recurring conditional language, not just fresh managerial doubt. In PayPal's 2024-filed 10-K, 173/232 (74.6%) uncertainty-containing sentences in Risk Factors also appear in its previous annual report after token normalization (minimum 12 words). Repeated warnings concern cyberattacks, fraud and business interruptions. This is direct evidence of retained risk disclosure in that case; it does not show that all uncertainty language across the sample is copied.

Q2. Are sentiment and uncertainty measuring different things? Pooled correlations are 0.841 (proportions) and 0.919 (TF-IDF); within 10-K they are 0.731/0.851, and within 10-Q 0.846/0.920. These are largely overlapping disclosure measures, not two independent confirmations. PayPal's 10-K filed 2024-02-08 (fiscal year ended 2023-12-31) illustrates a difference: Negative is 2.845% (89th percentile of 10-Ks), Uncertainty 1.790% (25th). Its discussion of transaction losses, credit-loss provisions and fraud protection raises Negative counts; some words describe ordinary business risks, rather than new bad news. Its four-day SPY excess return is -7.07%; annualized volatility falls from 37.94% to 25.14%. Those are descriptive case outcomes, not evidence that the wording caused them.

6. Trends, 2021-2025

Figure 1 separates report types and averages companies equally. Table 4 further adjusts for company identity and quarter-of-year seasonality. These corrections address changing form and company composition before interpreting trends.

Figure 1. Negative and uncertainty measures by filing quarter; VIX on right axes

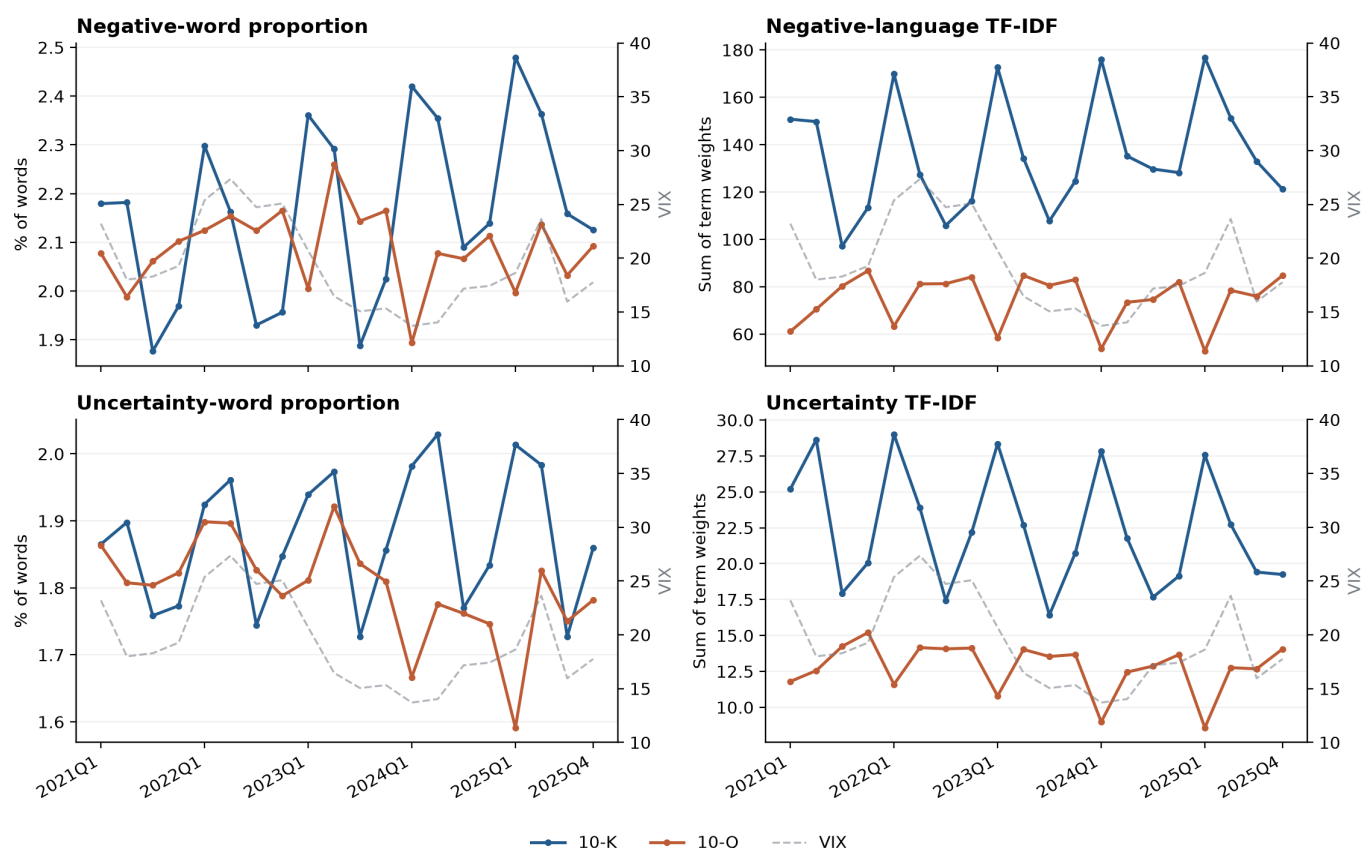


Table 4. Time-trend coefficient and t-statistics

Form / measure	Aggregate / year	OLS t	NW t	Within / year	Within t	Within p
10-K / Negative %	0.062	9.15	12.83	0.063	7.21	<0.001
10-K / Negative TF-IDF	2.892	3.27	5.85	2.630	3.78	<0.001
10-K / Uncertainty %	0.018	3.52	3.20	0.025	5.16	<0.001
10-K / Uncertainty TF-IDF	-0.194	-1.01	-1.51	-0.029	-0.26	0.797
10-Q / Negative %	-0.009	-0.77	-0.81	-0.007	-0.45	0.651
10-Q / Negative TF-IDF	-1.345	-1.85	-1.99	-1.519	-1.26	0.211
10-Q / Uncertainty %	-0.029	-2.88	-3.05	-0.020	-1.97	0.052
10-Q / Uncertainty TF-IDF	-0.516	-3.97	-4.64	-0.482	-3.46	<0.001

Dependent variable: the named language measure. Predictor: linear time in years. Proportional slopes are percentage points/year; TF-IDF slopes are term-weight units/year. OLS and NW t-statistics refer to the aggregate model; within t and p use company-clustered errors.

Q3. Did either measure trend over 2021-2025? I lead with the required within-company model because it removes stable differences between companies, not because it produces a preferred sign. Its similarity to the aggregate result supports the direction after that correction. Annual Negative rises 0.063 percentage points/year ($t=7.21$) and 2.630 TF-IDF units/year ($t=3.78$). Annual Uncertainty rises proportionally (0.025 points/year, $t=5.16$) but not in TF-IDF ($t=-0.26$). Quarterly Negative has no clear trend; quarterly Uncertainty falls -0.020 points/year ($t=-1.97$) and -0.482 TF-IDF units/year ($t=-3.46$). This supports some real within-company language change rather than a trend created entirely by composition; it does not prove management became more pessimistic or uncertain.

7. Uncertainty, volatility and returns

Table 5A. Does uncertainty predict subsequent volatility? Full sample

Dependent variable: log annualized volatility on [+4,+63]. Main independent variable: continuous uncertainty, computed separately as proportion or TF-IDF. Columns distinguish whether prior volatility is omitted or included. All columns include size, dollar volume, prior excess return, report type, company and calendar-quarter effects.

Variable / statistic	Proportion / Without prior-vol control	Proportion / With prior-vol control	TF-IDF / Without prior-vol control	TF-IDF / With prior-vol control
Uncertainty coefficient	0.0229	0.0005	0.0035	0.0014
Uncertainty clustered SE	0.0299	0.0250	0.0027	0.0019
Prior-volatility coefficient	-	0.3698	-	0.3682
Prior-volatility clustered SE	-	0.0964	-	0.0956
Uncertainty p-value	0.446	0.984	0.193	0.443
Filings / companies	1,536 / 91	1,536 / 91	1,536 / 91	1,536 / 91

Proportion coefficients are per 1 percentage point of all words; TF-IDF coefficients are per 1 term-weight unit. Standard errors cluster by company.

Q4. Does uncertainty language predict volatility? Adding prior volatility changes the proportional coefficient from 0.0229 to 0.0005 log points per percentage point, and TF-IDF from 0.0035 to 0.0014 log points per unit. The declines are 97.8% and 59.7%, respectively. The added control accounts for pre-existing volatility correlated with the language score. Both earlier estimates were already imprecise with the other controls and fixed effects; neither controlled estimate establishes additional predictive content.

Table 6. Does negative language predict four-day excess returns?

Dependent variable: stock-minus-SPY buy-and-hold return on [0,+3], in percentage points. Main independent variable: continuous Negative proportion or Negative TF-IDF. All Table 5 controls and log prior volatility are included. Proportion effects are per 1 percentage-point increase; TF-IDF effects are per 1 unit.

Negative-language predictor	Coefficient	Clustered SE	t	p	80% MDE
Negative % (+1 pp)	-0.815	0.706	-1.15	0.252	1.997 pp
Negative TF-IDF (+1 unit)	-0.012	0.010	-1.18	0.239	0.028 pp

N=1,536 filings and 91 company clusters. Approximate 80%-power minimum detectable effect (MDE) = (t critical at 5%, two-sided + 0.842) x clustered SE: 1.997 return percentage points per 1 proportion point and 0.028 return percentage points per TF-IDF unit. These are precision diagnostics for detecting effects, not measured test power.

Q6. Which of your results do you believe? I have the most confidence in the direction of the annual-report Negative trend within this selected sample. Proportion and TF-IDF both rise inside companies, with t-statistics of 7.21 and 3.78, and the aggregate estimates point the same way. This supports a change in filing language among surviving ARK issuers. It does not establish that their underlying businesses became more negative, because dictionary counts also capture recurring legal and accounting language.

I do not believe the evidence supports a general rise in Uncertainty. The annual proportion rises but annual TF-IDF does not, whereas both quarterly estimates fall; the measures are reacting differently to common hedging words and less common terms. For volatility, I believe the strong persistence in volatility and the absence of clear incremental information in the pooled language scores: after prior volatility is added, both uncertainty estimates are imprecise, with 95% intervals of [-0.049, 0.050] log points per proportion point and [-0.002, 0.005] per TF-IDF unit. The isolated controlled 10-K TF-IDF p-value in section 8 is exploratory because it is one of several form/weight specifications and no formal interaction test shows that 10-K and 10-Q effects differ. The return coefficients are negative, but their confidence intervals include zero and the MDEs are large. I therefore treat the return result as inconclusive, rather than evidence of no effect.

8. 10-K versus 10-Q

Table 5B. Uncertainty and subsequent volatility, by filing type. Table 5A estimates one common slope from all filings; Table 5B estimates the same model separately for 10-K and 10-Q. Every row uses Uncertainty; headings state the weighting and prior-volatility choice. Other controls are unchanged, and IDF is recomputed within each form.

Form	Uncertainty weighting	Without prior-vol control: coefficient / p	With prior-vol control: coefficient / p	N
10-K	Proportion (+1 pp)	-0.126 / 0.819	-0.132 / 0.796	381
10-K	TF-IDF (+1 unit)	0.038 / 0.062	0.033 / 0.041	381
10-Q	Proportion (+1 pp)	0.080 / 0.151	0.014 / 0.714	1,155
10-Q	TF-IDF (+1 unit)	0.008 / 0.062	0.003 / 0.280	1,155

Q5. Do 10-Qs behave like 10-Ks? The forms do not behave identically. Median length is 22,002 words for 10-Q versus 61,691 for 10-K. Negative-proportion SD is 0.940 versus 0.413 percentage points; Uncertainty SD is 0.582 versus 0.241. A shorter denominator can make a proportion more sensitive to a few changed words; repeated templates can instead suppress within-company variation. The observed overall variance is higher for 10-Q, but length alone does not explain it.

Annual reports contain a fuller discussion of business risks, so they are a plausible source of more company-specific text. That is a hypothesis, not a result of having more pages. Only annual Uncertainty TF-IDF is significant in the controlled form-specific volatility test; annual proportions and both quarterly measures are not. Different significance levels do not prove the forms differ. Table 4 similarly shows an annual Negative increase and a quarterly Uncertainty decline, not a common form-independent trend.

A 10-Q can arrive close to an earnings release, which may reveal information first or fall within the event window. Earnings-release timestamps are not matched here. Therefore, even a significant filing-date return association would not isolate the incremental effect of the 10-Q text.

9. Limitations

The 2026 holdings snapshot selects survivors; failed or exited former holdings are missing, and only 71/91 retained issuers appear in every filing year. Thus trends among survivors need not describe the historical ARK portfolio or the broader market. Only 20 filing quarters identify the aggregate time trend, so Newey-West inference is approximate in a short series.

The dictionary does not read context, negation or whether a sentence is new. The concentration of Uncertainty counts and the PayPal repetition result show that template language can drive scores. TF-IDF uses the full analysis corpus to calculate document frequencies, so it is retrospective rather than investable in real time. Daily event timing cannot separate filing text from simultaneous earnings news, and the multi-class share aggregation used for size is an approximation.

The current report includes aggregate and within-company trends; full-sample and form-specific volatility models with and without prior volatility; and SPY and ARKK return benchmarks. Multiple related tests raise the chance of an isolated small p-value. With ARKK, Negative coefficients are -0.602 per proportion point ($p=0.368$) and -0.007 per TF-IDF unit ($p=0.445$); the imprecise pooled return conclusion persists.

Earlier exploratory work also ran no-control, size-only, prior-volatility-only and size-plus-volatility outcome models, an issuer/time-effects sensitivity, a shell-company screen, a temporary sample without the \$3 cutoff, and a return-quintile plot. The September 7 instructions superseded those choices. They are disclosed in AI_USE.md and were not used to choose the reported conclusion. The PayPal repetition check remains one case, not a corpus-wide duplication estimate.

10. What I would do next

The single change that would most improve the test is reconstructing point-in-time ARK membership from dated holdings archives, then including each company only during periods when an ARK fund actually held it. This would restore firms later sold, delisted or failed and directly test whether the conclusions survive outside today's selected survivors. The cost is reconciling ticker, name and CIK changes and recovering filings plus delisting-adjusted prices; missing archives may also shrink the usable sample.

Sources: Loughran and McDonald (2011), Journal of Finance 66, 35-65; instructor assignment sheet and repository instructions; SEC EDGAR; Yahoo Finance. Saved calculations, full regression coefficients and filing-text comparisons are in the notebook. AI assistance is disclosed in AI_USE.md.